Sum and Difference Formulas

ANSWER KEY

Section 1: Sine Sum and Difference Identities

1 State the formula for $\sin(A+B)$.

a. $\sin A\cos B+\cos A\sin B$ ✓

b. $\sin A\cos B-\cos A\sin B$

c. $\cos A\cos B-\sin A\sin B$

d. $\sin A+\sin B$

The sine sum formula is $\sin(A+B)=\sin A\cos B+\cos A\sin B$.

2 Find the exact value of $\sin(75^\circ)$ using $75^\circ=45^\circ+30^\circ$.

a. $\frac{\sqrt{6}+\sqrt{2}}{4}$ ✓

b. $\frac{\sqrt{6}-\sqrt{2}}{4}$

c. $\frac{\sqrt{3}+1}{2}$

d. $\frac{\sqrt{2}+1}{2}$

$\sin(45^\circ)\cos(30^\circ)+\cos(45^\circ)\sin(30^\circ)=\left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right)+\left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)=\frac{\sqrt{6}+\sqrt{2}}{4}$.

3 Simplify $\sin(x+\pi)$.

a. $-\sin x$ ✓

b. $\sin x$

c. $-\cos x$

d. $\cos x$

$\sin x\cos\pi+\cos x\sin\pi=\sin x(-1)+\cos x(0)=-\sin x$.

Section 2: Cosine Sum and Difference Identities

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4 State the formula for $\cos(A - B)$.

a. $\cos A \cos B + \sin A \sin B$ ✓

b. $\cos A \cos B - \sin A \sin B$

c. $\sin A \sin B - \cos A \cos B$

d. $\cos A - \cos B$

The cosine difference formula is $\cos(A - B) = \cos A \cos B + \sin A \sin B$.

5 Find the exact value of $\cos(15^\circ)$ using $15^\circ = 45^\circ - 30^\circ$.

a. $\frac{\sqrt{6} + \sqrt{2}}{4}$ ✓

b. $\frac{\sqrt{6} - \sqrt{2}}{4}$

c. $\frac{\sqrt{3} - 1}{4}$

d. $\frac{\sqrt{2} + 1}{4}$

$\cos(45^\circ)\cos(30^\circ) + \sin(45^\circ)\sin(30^\circ) = \frac{\sqrt{2}}{2}\frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2}\frac{1}{2} = \frac{\sqrt{6} + \sqrt{2}}{4}$.

6 Simplify $\cos(x - \pi/2)$.

a. $\sin x$ ✓

b. $-\sin x$

c. $\cos x$

d. $-\cos x$

$\cos x \cos(\pi/2) + \sin x \sin(\pi/2) = \cos x(0) + \sin x(1) = \sin x$.

Section 3: Tangent Sum and Difference Identities

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7 State the formula for $\tan(A+B)$.

a. $\frac{\tan A+\tan B}{1-\tan A\tan B}$ ✓

b. $\frac{\tan A-\tan B}{1+\tan A\tan B}$

c. $\frac{\tan A+\tan B}{1+\tan A\tan B}$

d. $\tan A+\tan B$

The tangent sum formula is $\tan(A+B)=\frac{\tan A+\tan B}{1-\tan A\tan B}$.

8 If $\tan A=1/2$ and $\tan B=1/3$, find $\tan(A+B)$.

a. 1 ✓

b. $5/6$

c. $5/7$

d. $2/3$

$\frac{1/2+1/3}{1-(1/2)(1/3)}=\frac{5/6}{1-1/6}=\frac{5/6}{5/6}=1$.

9 Evaluate $\frac{\tan(80^\circ)-\tan(20^\circ)}{1+\tan(80^\circ)\tan(20^\circ)}$.

a. $\sqrt{3}$ ✓

b. $1/\sqrt{3}$

c. 1

d. 0

This matches the formula for $\tan(80^\circ-20^\circ)=\tan(60^\circ)=\sqrt{3}$.

Section 4: Compound Angle Problems with Given Ratios

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10 Given $\sin\alpha = 3/5$ (Quadrant I) and $\cos\beta = -5/13$ (Quadrant II), find $\sin(\alpha+\beta)$.

a. $33/65$ ✓

b. $-33/65$

c. $63/65$

d. $-63/65$

$\cos\alpha = 4/5$. $\sin\beta = \sqrt{1-(-5/13)^2} = 12/13$.
$\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta = (3/5)(-5/13) + (4/5)(12/13) = -15/65 + 48/65 = 33/65$.

11 For the same angles ($\cos\alpha = 4/5, \sin\alpha = 3/5, \cos\beta = -5/13, \sin\beta = 12/13$), find $\cos(\alpha+\beta)$.

a. $-56/65$ ✓

b. $56/65$

c. $-16/65$

d. $16/65$

$\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta = (4/5)(-5/13) - (3/5)(12/13) = -20/65 - 36/65 = -56/65$.

12 In which quadrant does $\alpha+\beta$ lie in the previous problems?

a. **Quadrant II** ✓

b. Quadrant I

c. Quadrant III

d. Quadrant IV

Since $\cos(\alpha+\beta) = -56/65 < 0$ and $\sin(\alpha+\beta) = 33/65 > 0$, the angle $\alpha+\beta$ terminates in Quadrant II.

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